

投稿系统单独提交项：Graphical Abstract 说明文字 + Highlights

Graphical Abstract —Caption

Delta Dependency (DD) framework with five-layer orthogonal validation. DD measures the shift in mean CERES dependency of a paralog gene between driver wild-type and mutant cell lines (manuscript Eq. 1). Top: from DepMap 26Q1 data (1,208 cell lines; 66,595 HGNC paralog pairs; 12 gold-standard pairs), DD nominates priority paralog-SL candidates without any training and compares favorably with published methods (AUROC 0.794 vs best published CV3 result 0.790; 1.000 with a $\geq 30\%$ sequence-identity filter). Bottom: orthogonal validation across genomic (23 solid tumor lineages; AUROC > 0.7 in 9 of 17 evaluable lineages), proteomic (seven CPTAC cohorts, $n = 672$), pharmacologic (1,482 PRISM compounds; 553 selective associations), clinical-stratification (MSI status and mutation type), and structural (sequence features, domain architecture, PROTAC suitability) layers. Fully reproducible from public data with no GPU requirement.

Highlights (每条 ≤ 85 字符)

1. A single-subtraction metric (Delta Dependency) predicts paralog synthetic lethality
2. DD alone (AUROC 0.736) outperforms four multi-feature classifiers in cross-validation
3. Protein-level paralog compensation is confirmed across seven CPTAC cohorts ($n = 672$)
4. Therapeutic window analysis nominates ARID1A-ARID1B as the leading SL candidate
5. Every quantitative claim is recomputable from raw data with a single command